# DPUTTSServerPico

Electronic speech synthesis („text-to-speech“) with the use of SVox Pico speech synthesis engine.

## 1. Static parameters:

| | |
|---|---|
| *Path* | Path (below the SILAB installation folder) where the language resource files are located.<br>default: *lib/tts/* |
| *TaFileName* | Filename of the „Textual analysis“ resource file containing textual information about the spoken language.It is searched below the given Path.<br>default: *de-DE_ta.bin* |
| *SgFileName* | Filename of the „Signal generation“ resource file containing voice samples of the spoken language. It is searched below the given Path.<br>default: *de-DE_gl0_sg.bin* |
| *Port* | TCP port used for communication between the different speech synthesis DPUs.<br>default: 0 which means the port given in the SILAB network configuration (*sys.par*) is used. |
| *PicoMemSizeKB* | Size of the system memory (in kilobytes) that is available for the Pico engine. Should be at least two times the size of all loaded resource files combined.<br>default: 4096 kB |

## 2. Dependencies:

DPUTTSServerPico must be used together with DPUTTSSNDX and DPUTTSSCNX.

## 3. Functional principle:

**DPUTTSServerPico** opens a TCP port and waits for incoming connections.

As soon as a connection has been established, all data is read line by line, interpreted as UTF8 coded strings then synthesised using the Pico engine and sent back to the client over the TCP connection.

**DPUTTSClientSNDX** can establish such connections, recieves the synthesised audio files over the network connection and plays them back over OpenAL. Using this approach, the audio source can be positioned in the 3D SILAB world.

**DPUTTSSCNX** may be loaded additionally to *DPUTTSClientSNDX* and allows to trigger playback via events of the SCNX database. (See documentation for details.)

```
# synthesize sounds
DPUTTSServerPico TTSPico
{
	Computer = {TTSP};
	Index = 20;
	TreeviewGroup = "Sound";
};

# play back synthesized sound
DPUTTSClientSNDX TTSClientPico
{
	Computer = {TTSP};
	Index = 50;
	TreeviewGroup = "Sound";
	ServerIP = "TTSP";
};

# and support hedgehogs
DPUTTSSCNX TTSSCNX
{
	Computer = { TTSP};
	Index = 51;
	TreeviewGroup = "Sound";
};
```

## 4. Pico markup commands:

Pico markup commands can be added to the text to modify speech output.

A complete description of the Pico markup commands can be found in the SVox Pico documentation. Some important ones are:

| | |
|---|---|
| *<pitch level='nnn'> … </pitch>* | Sets the pitch of the voice to the given level. *nnn* may range from 50 to 200 where 100 marks the normal pitch level. Only text between the opening and closing pitch tag is affected. |
| *<speed level='nnn'> … </speed>* | Sets the speed of the voice to the given level. *nnn* may range from 50 to 200 where 100 marks the normal speech speed. Only text between the opening and closing speed tag is affected. |
| *<volume level='nnn'> … </volume>* | Sets the volume to the given level. *nnn* may range from 50 to 200 where 100 marks the normal volume. Only text between the opening and closing volume tag is affected. |
| *<phoneme ph='abc'/>* | Outputs a word in phonetic script. This is useful in case the (uncommon) word is not pronounced correctly. *abc* is the SAMPA code of the word to speak. An introduction to SAMPA transcription is available on Wikipedia [at https://en.wikipedia.org/wiki/SAMPA_chart in |

English and at https://de.wikipedia.org/wiki/SAMPA-Transkribier-ungscodes in German]. A com-plete description of the supported phonetic codes for each language is available in the SVox Pico man-ual [available at https://salsa.debian.org/a11y-team/svox/blob/debian-sid/pico_resources/docs/SVOX_Pico_Man-ual.pdf].